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RV College of Engineering®
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Bengaluru - 560059, Karnataka, India

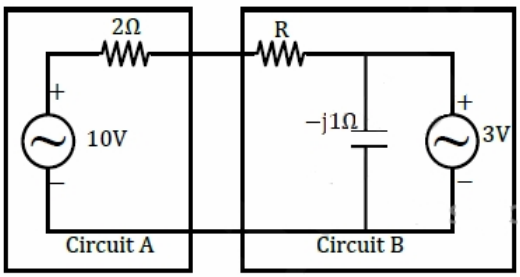
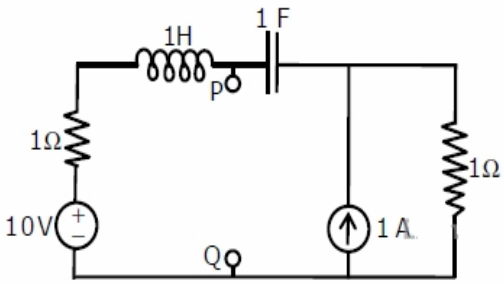
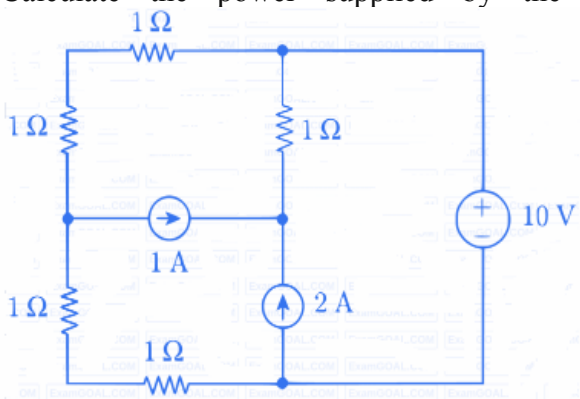
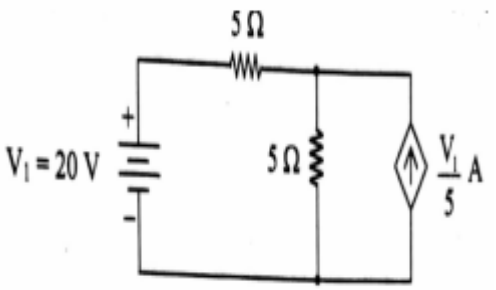
NBA Accredited (UG - 6 Years
PG - 3 Years)

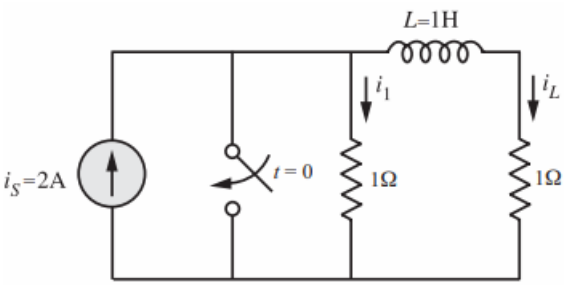
Department of Electronics & Communication Engineering

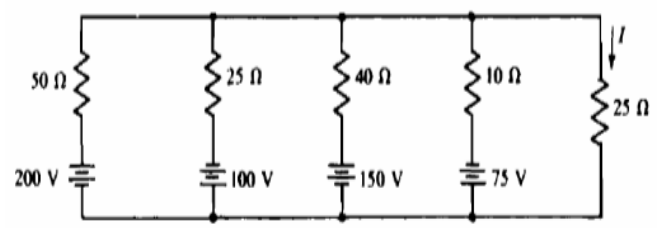
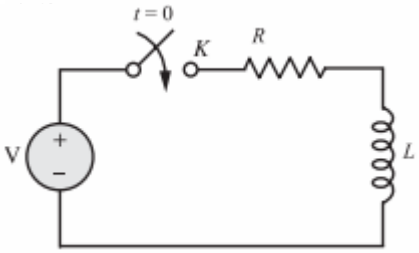
Date: 10/11/2025	**CIE -1**	**Max. Marks:60**
Semester: III	**UG**	**Duration: 2 Hrs**
Course: Network Analysis and Control Engineering		**Code:EC235AT**

NOTE: i) Quiz answers to be written in first two pages of answer booklet only
ii) Please adhere to the following guidelines:

1. Allowed Materials: Only prescribed Reference textbooks mentioned in the syllabus copy are permitted. Students must bring their own copy of the approved reference books mentioned in the syllabus copy.
2. No Sharing: Students are not allowed to share materials or communicate with each other during the CIE.
3. Electronic Devices: Use of electronic devices (e.g., laptops, tablets, smart watches etc) is not allowed during the CIE.
4. Handwritten, printed notes or any other photo copy materials are not allowed.
5. All answers should be written in the answer booklet provided by the Department.

S No	Part-A Quiz	M	BT	CO
1.	Assuming both the voltage sources are in phase, find the value of R for which maximum power is transferred from circuit A to circuit B shown in Figure 1.  Figure 1  Figure 2	2	3	1
2.	Find the Thevenin equivalent impedance Z_{th} between the nodes P and Q in the steady state for the circuit in Figure 2.	2	2	1
3.	Calculate the power supplied by the voltage source for the Figure 3  Figure 3  Figure 4	2	1	2

	A dependent current source is shown in Fig.4. The power due to dependent source is _____W.	2	2	2
5.	Find $i_1(0+)$ and $i_L(0+)$. The circuit is in steady state for $t < 0$.  Figure 5	2	1	2
S No	**Part-B Test**	**M**	**BT**	**CO**
1	1. For an AC circuit with frequency f=50Hz: • Mesh 1: R1=10Ω, L1=0.1H and voltage source V1=100∠45°V • Mesh 2: R2=8Ω, C2=100μF Meshes share a resistor R3=4Ω. (a) Write mesh equations in polar form. (b) Determine the current magnitude and phase for each mesh. (c) Find the voltage across the capacitor.	**10**	**3**	**1**
2	A voltage source of 15 V connects Node 1 and Node 2 directly and other elements: • R1=10Ω between Node 1 and ground • R2=20Ω between Node 2 and ground • R3=5Ω between Node 1 and Node 3 • Is=2A current source from Node 3 to ground (a) Identify the supernode and write equations. (b) Derive the matrix form of nodal equations. (c) Find voltages at Nodes 1, 2, and 3	**10**	**3**	**1**
3	A network contains: • R1=2Ω, R2=4Ω, R3=6Ω • Independent voltage source V=12V • A dependent current source of 0.2Vx, where Vx is the voltage across R1. The load resistor R_L=3Ω is connected between terminals A–B. (a) Find Thevenin's equivalent circuit across A–B, including the dependent source. (b) Compute I_{RL}. (c) Verify by converting to Norton's form. (d) Compare results if dependent source were removed and discuss the effect.	**10**	**3**	**1**

4	A DC source V=24V with internal resistance 6Ω and a current source 3A with parallel resistance 12Ω supply a load R_L. (a) Using Superposition, find equivalent voltage across the load. (b) Determine R_L for maximum power transfer. (c) Calculate the total power delivered under this condition	**8**	**3**	**1**
5a	Using Millman's theorem, calculate current through the load (1+j)Ω in the circuit the Figure 5a.   **Figure 5a** **Figure 5b**	**6**	**2**	**2**
5b	In the network shown in Figure 5b, K is closed at t=0 with zero current in the inductor. Find the values of i(0), di/dt, d^2i/d^2t at t=0+ , if R=8Ω,L=0.2H,V=12V in A/S and A/S^2.	**6**	**2**	**2**

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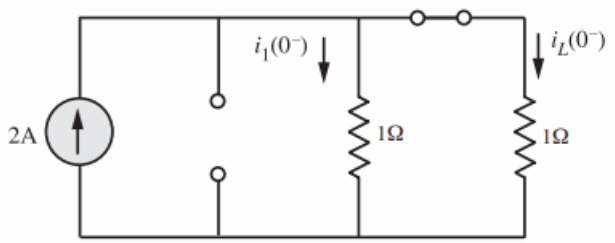
SCHEME & SOLUTIONS

S No	Solutions with Scheme	Marks
1.	0.8 Ω	2
2.	1Ω For DC source under steady state, inductor behaves as a short circuit and capacitor behaves as an open circuit.	2
3.	0W Applying Kirchhoff's voltage law in the outer loop, $-10 + (3+i)2 + (2+i)2 = 0$ Solving the above equation, we get $i = 0$ The power supplied by the voltage source is $P = Vi = 10 \times 0 = 0\,\mathrm{W}$	2

	KVL equation:1M P=80W 1M	2
5.	 $i_L(0^-) = \frac{2 \times 1}{1+1} = 1\text{A}$ $\quad i_L(0^+) = i_L(0^-) = 1\text{A}$ At $t = 0^-$, $i_1(0^-) = 2 - 1 = 1\text{A}$. Please note that the current in a resistor can change instantaneously. Since at $t = 0^+$, the switch is just closed, the voltage across R_1 will be equal to zero because of the switch being short circuited and hence, $i_1(0^+) = 0\text{A}$ Thus, the current in the resistor changes abruptly form 1A to 0A.	2
	PART B	

1

Given $f = 50\text{ Hz}, \omega = 2\pi f = 100\pi\text{ rad/s} \approx 314.159\text{ rad/s}.$

Compute element impedances:

- $Z_{L1} = j\omega L_1 = j(314.159)(0.1) = j31.4159\ \Omega.$
- $Z_{C2} = \dfrac{1}{j\omega C_2} = -j\frac{1}{\omega C_2} = -j\frac{1}{314.159\times100\times10^{-6}} = -j31.83099\ \Omega.$

KVL for Mesh 1 (clockwise, positive reference):

$$(R_1 + j\omega L_1 + R_3)I_1 - R_3 I_2 = V_1$$

KVL for Mesh 2:

$$-R_3 I_1 + (R_2 + Z_{C2} + R_3)I_2 = 0$$

Substitute numeric impedances:

$$(14 + j31.4159)\, I_1 - 4\, I_2 = 100\angle 45^\circ$$

$$-4\, I_1 + (12 - j31.83099)\, I_2 = 0$$

(b)

- $I_1 = 2.7531 - j1.0532\text{ A} = 2.95\angle -20.93^\circ\text{ A}.$
- $I_2 = 0.23008 + j0.25923\text{ A} = 0.3466\angle 48.41^\circ\text{ A}.$

(c)

Voltage across capacitor (phasor) $V_C = I_2 \cdot Z_{C2}.$

With $Z_{C2} = -j31.83099\ \Omega$ and the computed I_2:

$$V_C \approx 11.033\angle -41.59^\circ\text{ V}.$$

So the capacitor voltage magnitude is $\boxed{11.03\text{ V}}$ with phase -41.59°.

2

Node 1 and Node 2 form a *supernode*.

Relationship due to voltage source:

$$V_1 - V_2 = 15$$

Apply KCL to the supernode (Nodes 1 & 2 together)

Currents leaving the supernode:

- Through R_1: $\frac{V_1}{10}$
- Through R_2: $\frac{V_2}{20}$
- Through R_3: $\frac{V_1 - V_3}{5}$

Sum of currents leaving supernode = 0:

$$\frac{V_1}{10} + \frac{V_2}{20} + \frac{V_1 - V_3}{5} = 0$$

KCL at Node 3

Currents leaving Node 3:

- Through R_3: $\frac{V_3 - V_1}{5}$
- Through the current source: $2\ \mathrm{A}$ (downward, leaving node)

So:

$$\frac{V_3 - V_1}{5} + 2 = 0$$

(b)

$$\begin{bmatrix} 6 & 1 & -4 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -10 \\ 15 \end{bmatrix}$$

(c)

From Eq. (3): $V_2 = V_1 - 15$

From Eq. (2): $V_3 = V_1 - 10$

Substitute into Eq. (1):

$$6V_1 + (V_1 - 15) - 4(V_1 - 10) = 0$$

Simplify:

$$6V_1 + V_1 - 15 - 4V_1 + 40 = 0$$

$$3V_1 + 25 = 0$$

$$V_1 = -\frac{25}{3} = -8.33\,\text{V}$$

Then:

$$V_2 = V_1 - 15 = -8.33 - 15 = -23.33\,\text{V}$$

$$V_3 = V_1 - 10 = -8.33 - 10 = -18.33\,\text{V}$$

3

Step 1: Find Open-Circuit Voltage (Vth)

We remove R_L.

Let V_A be open-circuit voltage at terminal A (i.e. $V_{th} = V_A$).

By nodal equations:

At **Node 1** (connected to R_1, R_2, and dependent source):

$$\frac{V_1 - 12}{R_1} + \frac{V_1 - V_A}{R_2} + 0.2(12 - V_1) = 0$$

At **Node A:**

$$\frac{V_A - V_1}{R_2} + \frac{V_A}{R_3} = 0$$

Solving gives:

$$V_1 = 9.0\,\text{V}, \quad V_A = 5.4\,\text{V}$$

$$V_{TH} = 5.4\,\text{V}$$

(b)

Step 2: Find Thevenin Resistance (Rth)

Turn **off the independent source (12V → short circuit)** but keep the dependent source active.

Apply a 1 V test source at terminal A (to ground) and find resulting current I_{test}.

Equation at Node 1:

$$\frac{V_1 - 0}{R_1} + \frac{V_1 - 1}{R_2} + 0.2(0 - V_1) = 0$$

$\rightarrow V_1 = 0.4545\text{V}$

Current into the network from the test source:

$$I_{test} = \frac{1 - V_1}{R_2} + \frac{1}{R_3} = 0.3030\text{A}$$

So,

$$R_{TH} = \frac{V_{test}}{I_{test}} = \frac{1}{0.3030} = 3.30\ \Omega$$

$$\boxed{V_{TH} = 5.4\ \text{V}, \quad R_{TH} = 3.3\ \Omega}$$

With $R_L = 3\ \Omega$:

$$I_{RL} = \frac{V_{TH}}{R_{TH} + R_L} = \frac{5.4}{3.3 + 3} = 0.855\ \text{A}$$

$$V_{RL} = I_{RL} \times R_L = 0.855 \times 3 = 2.57\text{V}$$

(c)

Norton current:

$$I_N = \frac{V_{TH}}{R_{TH}} = \frac{5.4}{3.3} = 1.64\ \text{A}$$

and $R_N = R_{TH} = 3.3\Omega$.

Load current check:

$$I_{RL} = I_N \frac{R_N}{R_N + R_L} = 1.64\frac{3.3}{6.3} = 0.86A$$

(d) Effect of Removing the Dependent Source

If the dependent source (0.2 Vx) is **removed**:

Recalculate:

V_{TH}: Without dependent source, KCL simplifies → $V_{TH} = 4.8V$.
R_{TH}: Without dependent source, $R_{TH} = 4\Omega$.

Then,

$$I_{RL} = \frac{4.8}{4+3} = 0.686A$$

4

Case 1: Voltage source active, current source turned off

To "turn off" a **current source**, we open-circuit it.

So we have:

- 24 V source with series 6 Ω,
- in parallel with 12 Ω (since the current source branch is open).

Voltage across 12 Ω (and across output) is given by voltage division:

$$V_1 = 24 \times \frac{12}{6+12} = 24 \times \frac{12}{18} = 16\,\text{V}$$

So, due to the voltage source alone, $V_1 = 16\,\text{V}$.

Case 2: Current source active, voltage source turned off

To "turn off" a **voltage source**, we short-circuit it.

So now:

- 3 A source in parallel with 12 Ω,
- and also in parallel with 6 Ω (from the shorted voltage source branch).

We need voltage across this parallel combination.

First, find **equivalent resistance seen by current source**:

$$R_{eq} = 6 \parallel 12 = \frac{6 \times 12}{6+12} = 4\,\Omega$$

Voltage across them (from current source branch):

$$V_2 = I \times R_{eq} = 3 \times 4 = 12\,\text{V}$$

$$V_{TH} = V_1 + V_2 = 16 + 12 = 28\,\text{V}$$

(b)

Step 2: Find Thevenin Resistance

To find R_{TH}:

- **Turn off all independent sources:**
  - Voltage source → short circuit.
  - Current source → open circuit.

Now only resistors $6\,\Omega$ and $12\,\Omega$ are left in parallel.

$$R_{TH} = 6 \parallel 12 = \frac{6 \times 12}{6 + 12} = 4\,\Omega$$

For Maximum Power Transfer

$$R_L = R_{TH} = 4\,\Omega$$

(c)

At maximum power transfer:

$$I_L = \frac{V_{TH}}{R_{TH} + R_L} = \frac{28}{4 + 4} = 3.5\,\text{A}$$

$$P_{L,\max} = I_L^2 R_L = (3.5)^2 \times 4 = 12.25 \times 4 = 49\,\text{W}$$

5a

$$V_M = \frac{(1/50)(200) + (1/25)(-100) + (1/40)(150) + (1/10)(-75)}{1/50 + 1/25 + 1/40 + 1/10} = -20.27\ \text{V}$$

$$R_M = \frac{1}{1/50 + 1/25 + 1/40 + 1/10} = 5.41\ \Omega$$

$$I = \frac{V_M}{R_M + R} = \frac{-20.27}{5.41 + 25} = -0.667\ \text{A}$$

Formula : 2M

Soultion :4M

6

5b	KVL Equation and diagram after switching : 3M I(0+)=10 A 1M di/dt=60 A/sec 1M d^2i/d^2t= -2400 A/Sec^2 1M	6

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SCRUTINY & EVALUATION OF CIE QUESTION PAPER

Declaration by the Course handling faculties:

As a teaching faculty of the course Network Analysis and Control Engineering with code: EC235AT, We hereby confirm that the question paper with Scheme and Solutions is thoroughly reviewed and we ensure that it adheres to the following criteria:

Sl. No	Criteria	Yes/No

1	The question paper adequately covers the prescribed syllabus contents.	Y
2	The question paper is in line with the recommended pattern, taking into consideration the structure and format suitable for the Continuous Internal Evaluation.	Y
3	The question paper is designed to align with the Revised Bloom's Taxonomy, encompassing various levels of cognitive skills such as remembering, understanding, applying, analyzing, evaluating, and creating.	Y
4	The question paper is aligned with the defined course outcomes, ensuring that it effectively assesses the knowledge and skills acquired during the course.	Y
5	Course handling faculty (As applicable) are responsible for preparing the question paper, scheme, and solution have unanimously agreed to utilize this question paper for conducting the Continuous Internal Evaluation.	Y
6	The question paper, Scheme and Solution has been submitted to the Test coordinators within the designated timeframe to ensure the smooth conduction of Continuous Internal	Y

Course handling Faculties:

Name: **Signature**

1. Dr. Usharani K R
2. Dr. Veena Devi
3. Prof. Neeta B M
4. Prof. Sujata P M

To be filled by the Scrutinizer:

Sl. No	Rubrics	Points	
		Max	**Awarded**
1	Timely submission of the question paper along with the scheme & solution	10	
2	Heterogeneous nature of QP with respect to BTs and Cos	10	
3	Format with proper entry of all particulars including test, course name, code, date, max marks, BT CO table, efficient use of paper (proper spacing, figures)	10	
4	No handwritten data or diagrams	10	
5	Scheme & Solution in the format	10	
	Total points	**50**	

Any other comments by the scrutinizer :

Note: Course coordinators to obtain scrutinizer's acceptance by incorporating all suggestions from scrutiny into the final versions of QP, Scheme, and Solutions.

All corrections suggested by the scrutinizer are incorporated and both the copies are re-submitted	**Signature of Course coordinator**
Accepted/Rejected	**Signature of Scrutinizer** **(Name:)**

Signature of HOD